

Carlos J. Malave

Summary

Accomplished, detail-oriented Senior Electrical Engineer with 9+ years of experience providing critical technical oversight, system engineering, and safety compliance for high-consequence environments, including U.S. Navy programs and industrial automation. Proven record managing hardware life-cycle plans, executing quality assurance audits, and developing engineering documentation. Have an extensive background in electrical and automated control systems with growing capabilities in Python development to deliver scalable solutions.

Technical Skills

- **Engineering & Systems:** Lifecycle Management, Configuration Management, Root Cause Analysis, Analog/Digital Circuit Design, Quality Assurance.
- **Defense & Maritime:** Submarine Safety Program (SUBSAFE), Deep Submergence Systems (DDS/SRDRS), Re-Entry Control (REC) Packages, Audits & Survey Compliance.
- **Automation & Hardware:** PLC Programming, Control Circuits, Industrial Automation, Test Equipment (Oscilloscopes, Multimeters, RF Signal Generators, Power Supplies).
- **Operating Systems & Environments:** Linux (Ubuntu), Windows.
- **Software & Programming:** GHDL, GTKWave, Python, C++, MATLAB, AutoCAD 2020, PTC Windchill, MultiSim, PSpice, Git/GitHub, HTML/CSS.

License

Engineering Intern, State of Florida, License #1100029256

Certificates

Intro to VHDL for FPGA and ASIC design, Udemy, 9.5 total hours.

Python Essential Training, LinkedIn Learning, 4 total hours.

Reliability-Centered Maintenance (RCM).

Professional Development

May 2024 - Present

- Took a planned career break to focus on advanced technical upskilling, independent engineering projects, and localized family/professional transitions.
 - Dedicated 500+ hours to mastering Python programming, Git/GitHub version control, and data structures.
 - **Python-Based Game Development & Software Design:**
 - **Connect Four (Algorithmic Logic):** Designed and programmed a classic grid-based strategy game using Python. Built custom game-state detection logic (matrix traversal) to dynamically scan for horizontal, vertical, and diagonal win conditions.
 - **Flappy Bird (Event-Driven Programming & Physics):** Engineered a real-time, event-driven arcade game with collision-detection algorithms, gravity simulation, dynamic obstacle generation, and real-time user-input handling.
 - **Snake Game (Object-Oriented Design & Coordinate Math):** Implemented an interactive game utilizing object-oriented programming (OOP) principles in Python. Programmed array-based data structures to track coordinate movements of the snake, growth logic, and collision algorithm.
 - **Full-Stack Basics:** Developing modern front-end web development practices utilizing clean HTML, CSS, and interactive structures.
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Work Experience

Senior Electrical Engineer at Submarine Maintenance Engineering, Planning and Procurement (SUBMEPP) at Kittery, ME

Oct 2023 - May 2024

- Engineered and provided technical oversight for electrical hull fittings and outboard cables for submarines, ensuring compliance with stringent U.S. Navy standards.
- Developed and optimized Class Maintenance Plans (CMPs), driving continuous improvement in hardware lifecycle efficiency, component reliability, and procedural accuracy.
- Reviewed and approved Maintenance Standards, ensuring technical accuracy and safety.
- Authored and maintained electrical test procedures, contributing to the reliability of critical submarine systems.
- Efficiently resolved test problem reports from maintenance activities, minimizing downtime and ensuring operational readiness.

Electrical Engineer at Portsmouth Naval Shipyard (PNS) at Kittery, ME

May 2017 - Oct 2023

- Delivered technical and engineering support for the complex electrical systems of the Dry Deck Shelter (DDS) and Submarine Rescue Diving Recompression System (SRDRS).
- Authored, maintained, and managed the configuration of technical manuals, maintenance procedures, and schematic drawings for Deep Submergence System (DSS) assets.
- Served as a key technical contributor during formal NAVSEA functional audits and safety certification surveys, consistently maintaining stringent quality benchmarks.
- Audited and validated over 70 SUBSAFE and non-SUBSAFE deficiency forms, re-entry control (REC) packages, and task group instructions (TGIs) to verify completed system maintenance and quality compliance.
- Consulted directly with project developers and localized maintenance crews to troubleshoot and resolve immediate technical bottlenecks during deep submergence overhauls.

Control Engineer at Accurate Solution and Design at Mayaguez, PR

Jun 2014 - Feb 2017

- Designed, programmed, and commissioned industrial PLC automation systems and variable speed drive control circuits for complex applications, including a custom industrial press for Hewlett-Packard.
- Engineered and configured robust control panels and automatic transfer switches (ATS) for critical backup infrastructure, including water distribution pump systems for the Puerto Rico Aqueduct and Sewer Authority (PRASA).
- Executed detailed engineering site audits to assess structural and electrical parameters for residential and commercial solar power integration.
- Utilized AutoCAD to draft detailed electrical single-line diagrams, schematic layouts, and physical connection blueprints for renewable energy installations.
- Supervised, scheduled, and led a team of five field technicians, managing project timelines and field operations.

Production Worker at Sweet Prince Gifts at Palm Bay, FL

(seasonal) Jul 2025 - Dec 2025

- Ensuring high-quality products are delivered to customers on time.
- Provide exceptional customer service and support to ensure customer satisfaction.

Education

University of Puerto Rico, Mayaguez Campus

Aug 2008 - May 2014

Bachelor of Science in Electrical Engineering - Magna Cum Laude

Engineering Capstone Projects

- **Metabolic Calorimetry Instrument (Lead Designer):** Spearheaded the architecture and construction of a metabolic rate measurement device. Designed and routed a custom PCB to interface oxygen, airflow, pressure, humidity, and temperature sensors with a central microprocessor; compiled thorough verification testing documentation.
- **3-Lead ECG Monitor:** Designed, simulated, and built a functional 3-lead electrocardiogram (ECG) monitor utilizing operational amplifiers and analog integrated circuits with active lead-off detection circuitry.